



Pre-Event Report

***Engineering report combining track analysis, car setup sensitivity,
driver comparison and race strategy ahead of round 1 of the
Virtual TCR Championship***

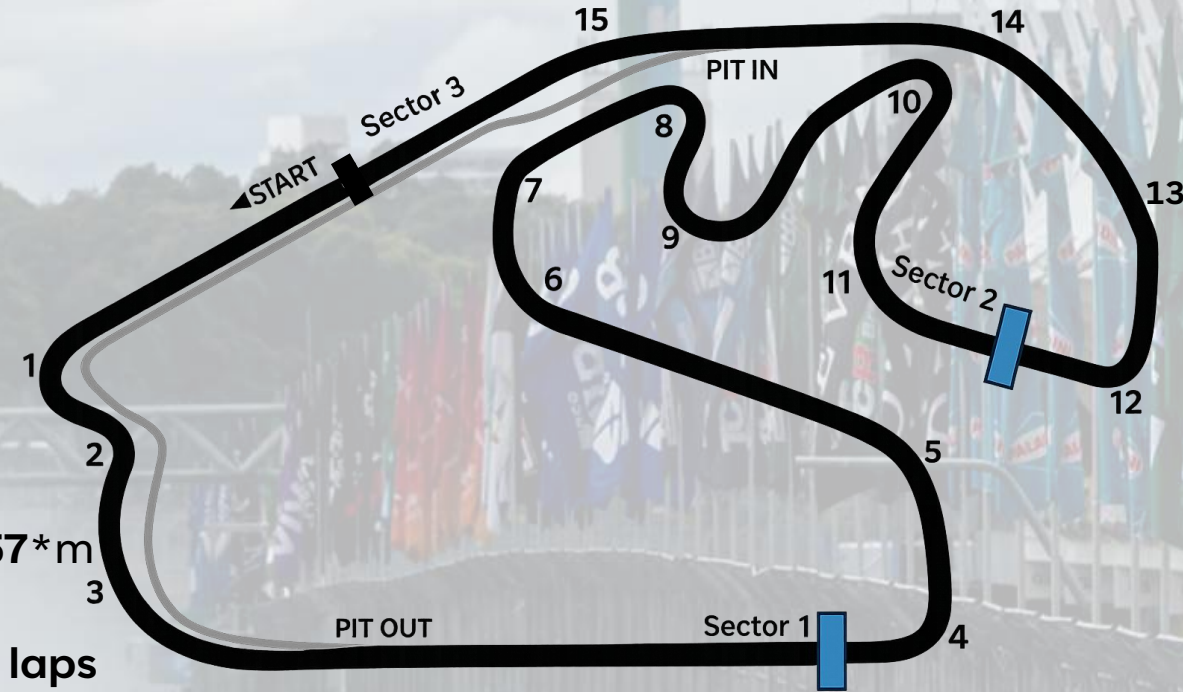
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Main metrics:

- Length: **4309** m, 2nd longest track on calendar
- Elevation: **43** m, 3rd highest on calendar
- Corners: **15** → Left: 10, Right: 5; Ratio → **2:1**
- Track Direction: **Anti** Clockwise
- Expected laptime (dry/wet): **1:42.615/1:53.109***
- Top speed (dry/wet): **241/229*** km/h (Str1)
- Average speed (dry/wet): **149/137*** km/h
- Minimum speed (dry/wet): **75/53*** km/h (T9)
- Distance from starting line to T1 braking (dry/wet) ≈ **151/157*** m
- Weekend format: **2X30min FP; 20min Q1; 10min Q2;**
25min + 1lap R1 ≈ 14-16 laps ; 30min + 1 lap R2 ≈ 18-20 laps



Track Characterization:

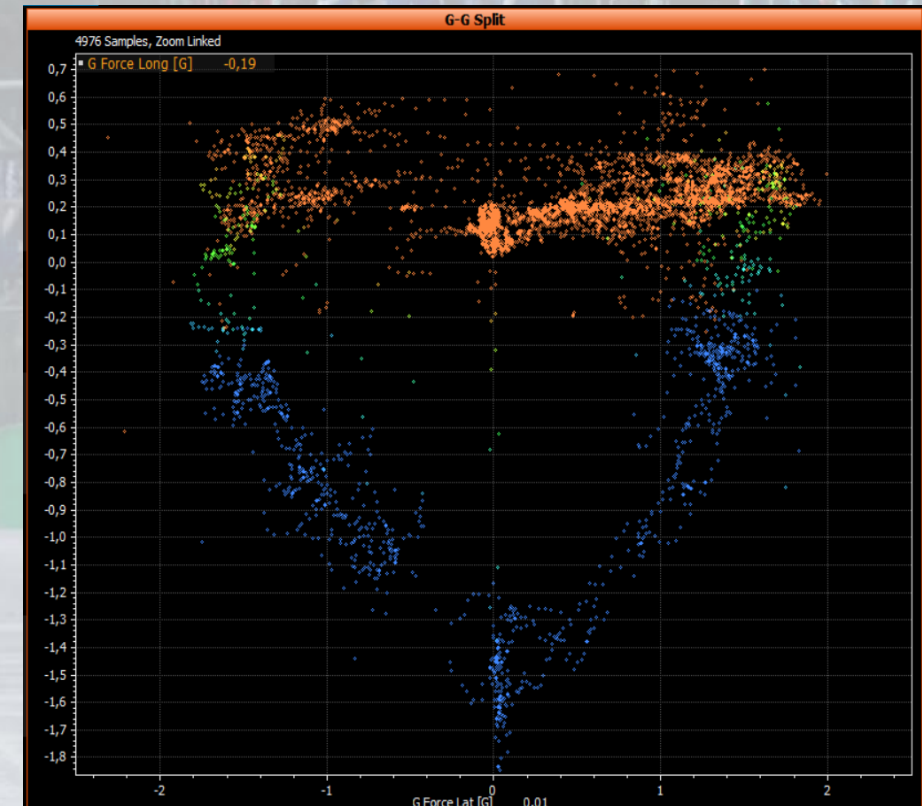
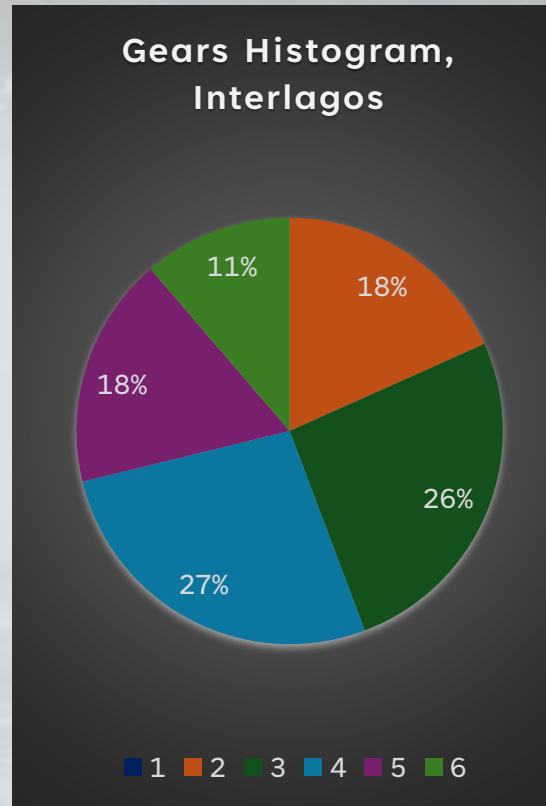
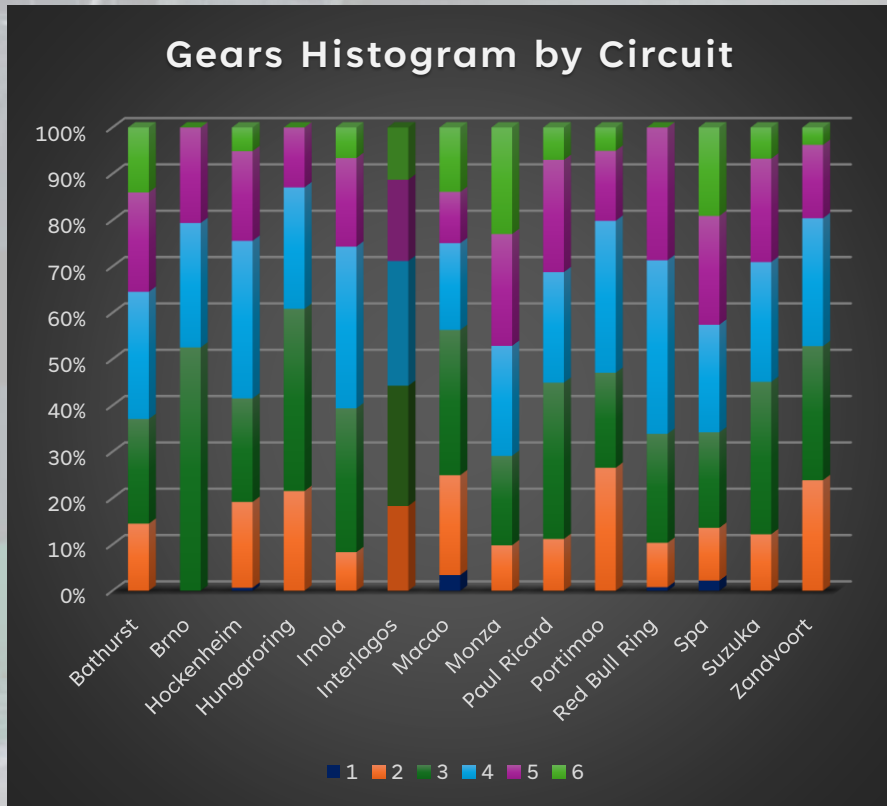
- Sectors 1 and 3 are fast and straight, whereas Sector 2 is the more technical section
- Aerodynamics matter on the main straight, but **mechanical grip is more important** due to the elevation changes and medium-speed corners
- The circuit was resurfaced in 2014, the asphalt is relatively aged, and both **brake and tyre wear tend to be high**

*Data from previous races and cross validated with MoTeC sources

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Key Performance Indicators:

- **Gears Histogram:** First gear is not used at this circuit, while **third gear** is the most frequently used, indicating predominantly **medium- to high-speed** corners. Overall, the **gear distribution** is **well balanced** across the lap. Compared to other tracks, it can be seen, that it does not have any irregularities.
- **Track characterization by G-G Split:** LH corners are **dominant** → **higher wear on the FR tyres**.

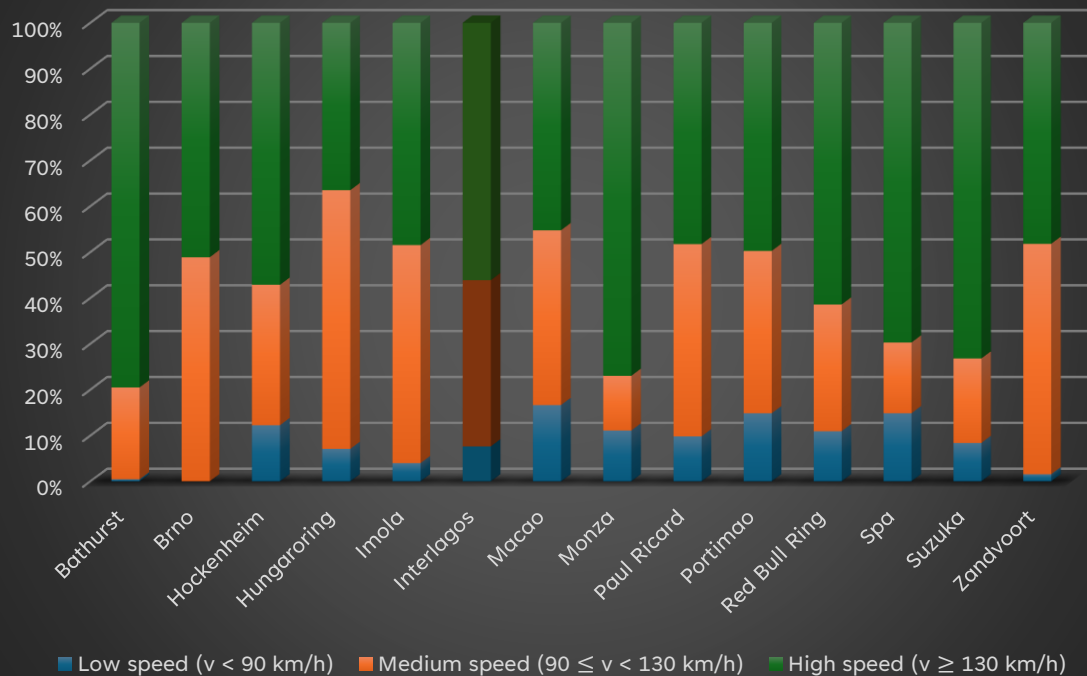


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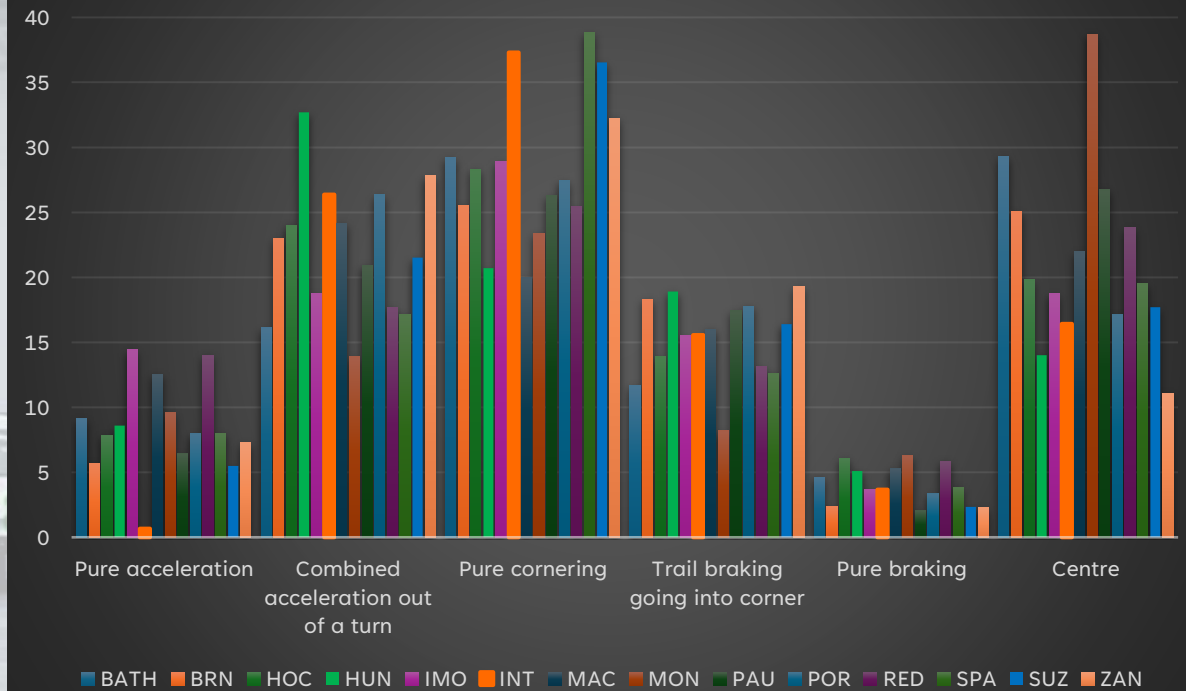
Key Performance Indicators:

- **Corner Speed Distribution by Circuit:** mainly defined by **medium- and high-speed corners**, without showing any extreme values compared to the rest of the calendar.
- **Track Characterization by G-G division:** the **lowest pure acceleration** share, **relatively high combined acceleration** out of corners, and a **strong pure cornering** component, **pointing to a flow-dominated layout** with sustained lateral load rather than long straight-line traction zones.

Corner Speed Distribution by Circuit

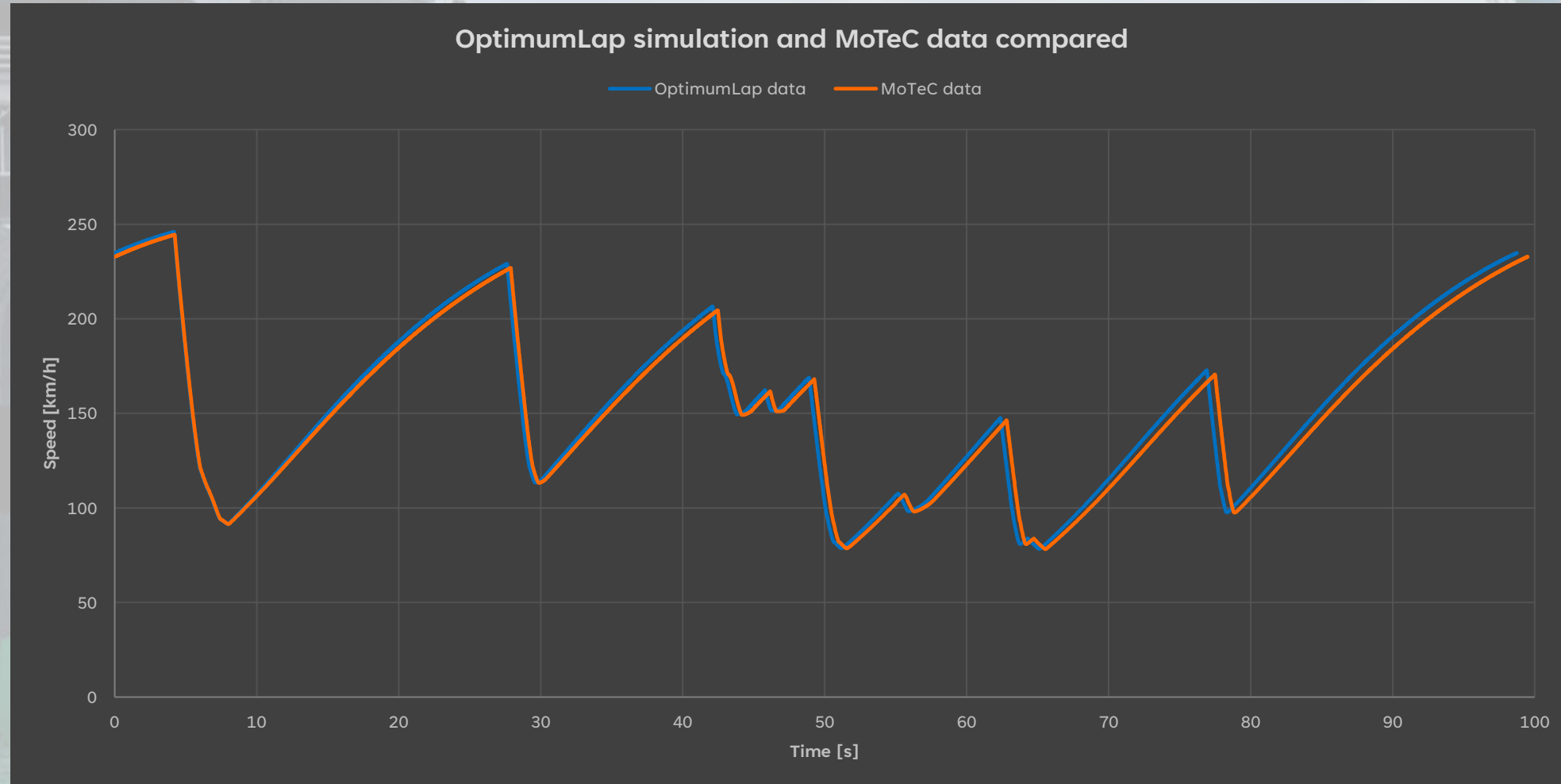


Track Characterization by G-G division



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Optimum Lap Simulation Validation:



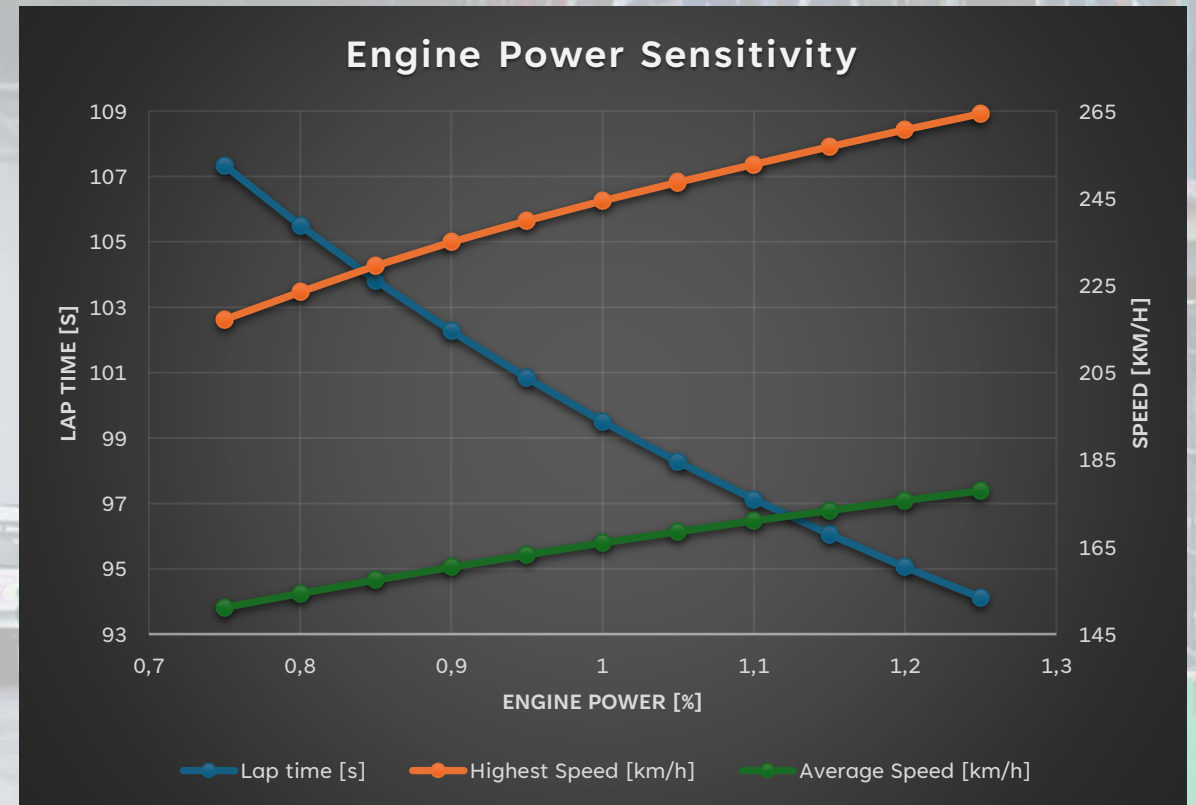
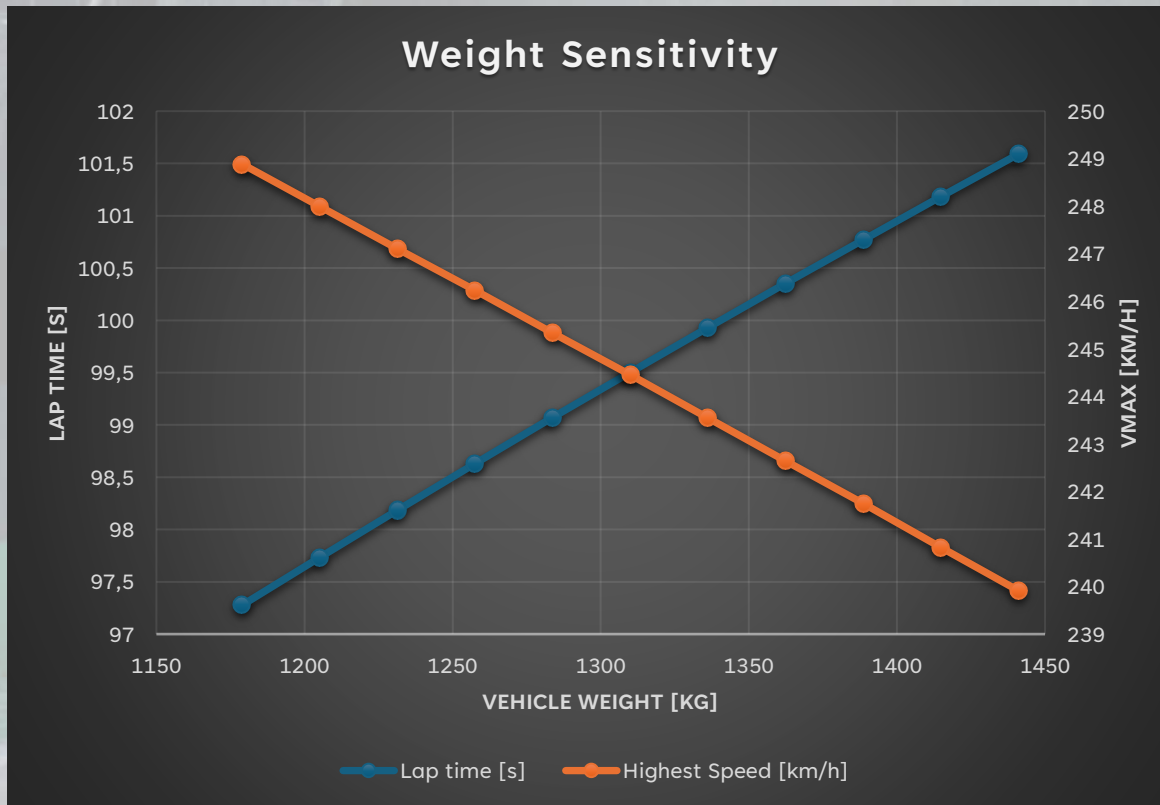
OptimumLap time	1:39,5
MoTeC best lap time	1:39,528
Diiference	0,28 s
Vehicle Mass	1310 kg
Max Torque	450 Nm @ 4000 rpm
Max Power	263 kW @ 6000 rpm

To acquire parameter sensivity values, the **OptimumLap data must be validated**. As shown in the chart, the two simulation data **closely match**, confirming **the model is accurate** for further analysis.

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Parameter Sensitivites:

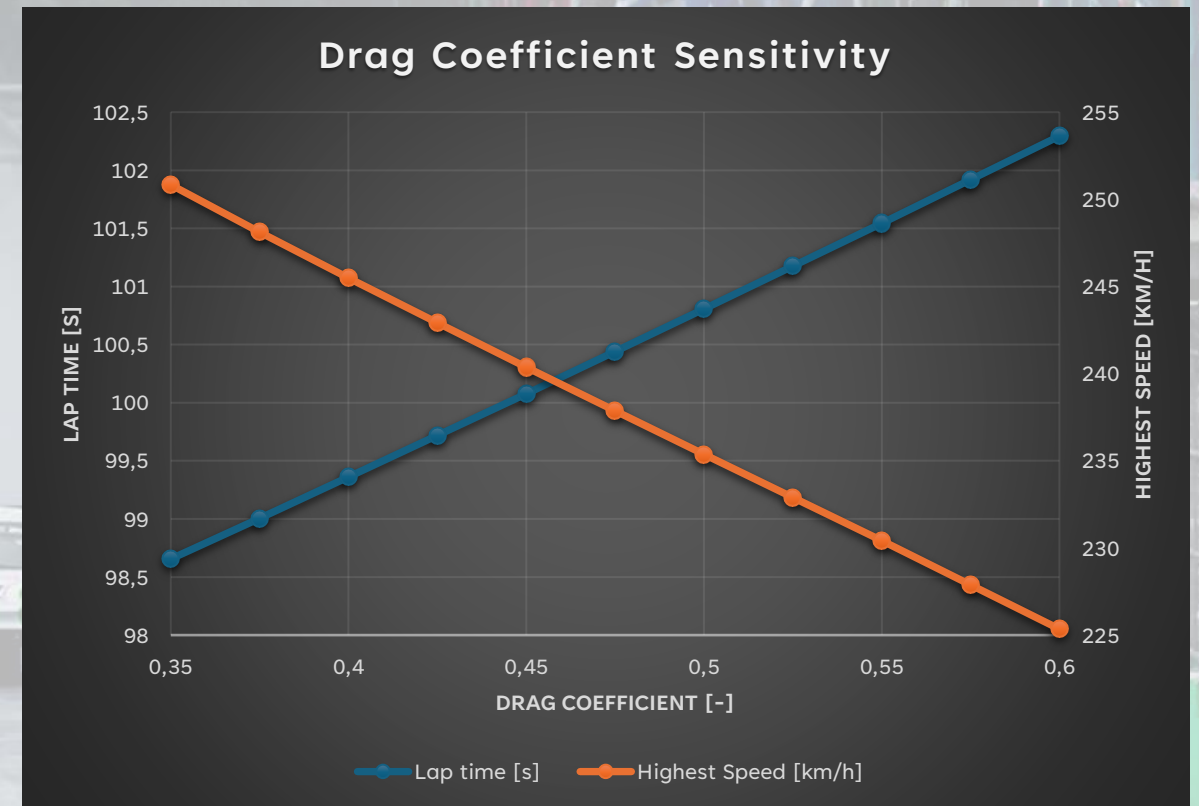
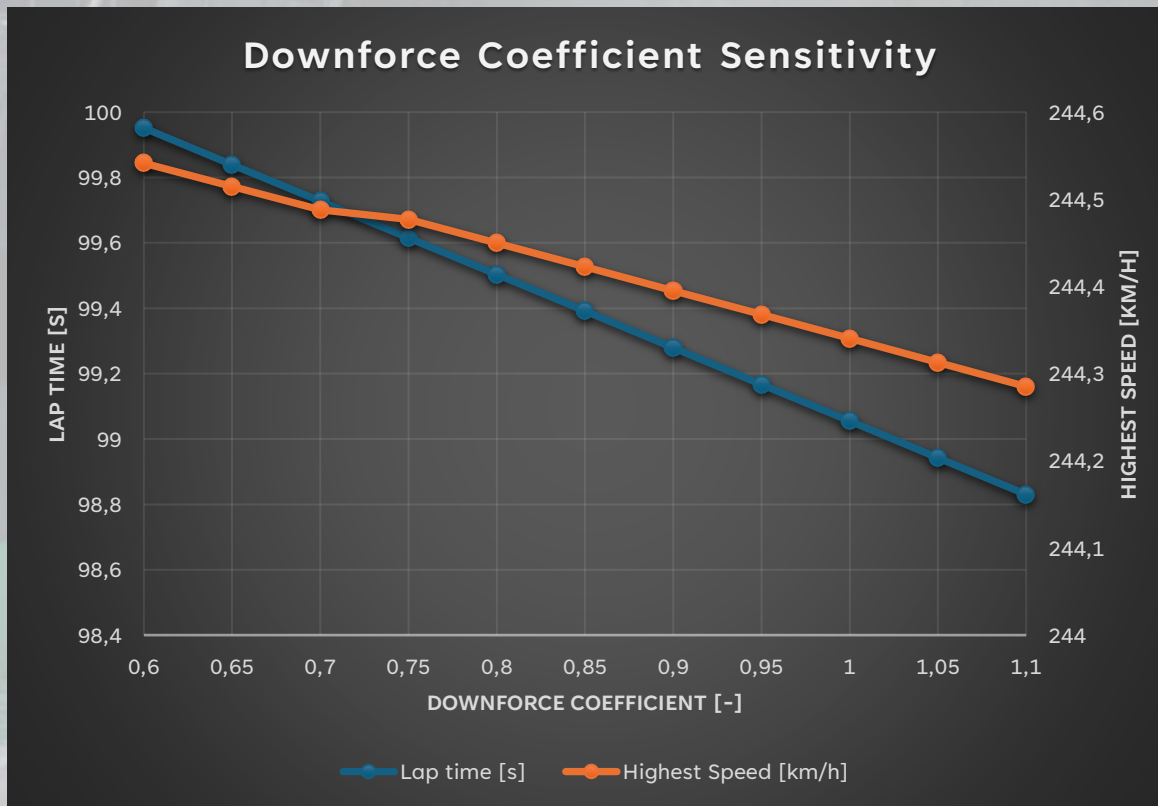
- **Weight Sensitivity:** In general, before a session, the car usually weights around 1310 kg with fluids and driver. **Vehicle mass is a key parameter influencing overall performance** at this circuit, as it directly affects **acceleration capability, braking distances, and cornering performance.**
- **Engine Power Sensitivity:** At Interlagos, the **combined acceleration out of corners is particularly high**, making **engine power a critical factor** for achieving **strong exits and overall lap performance.**



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Parameter Sensitivites:

- **Downforce Coefficient Sensitivity:** Higher downforce significantly improves lap times with minimal impact on top speed. On TCR cars this can be tuned via wing angle adjustments.
- **Drag Coefficient Sensitivity:** Increased drag severely penalizes both top speed and lap times. Optimized by balancing aerodynamic profile and rear wing pitch.

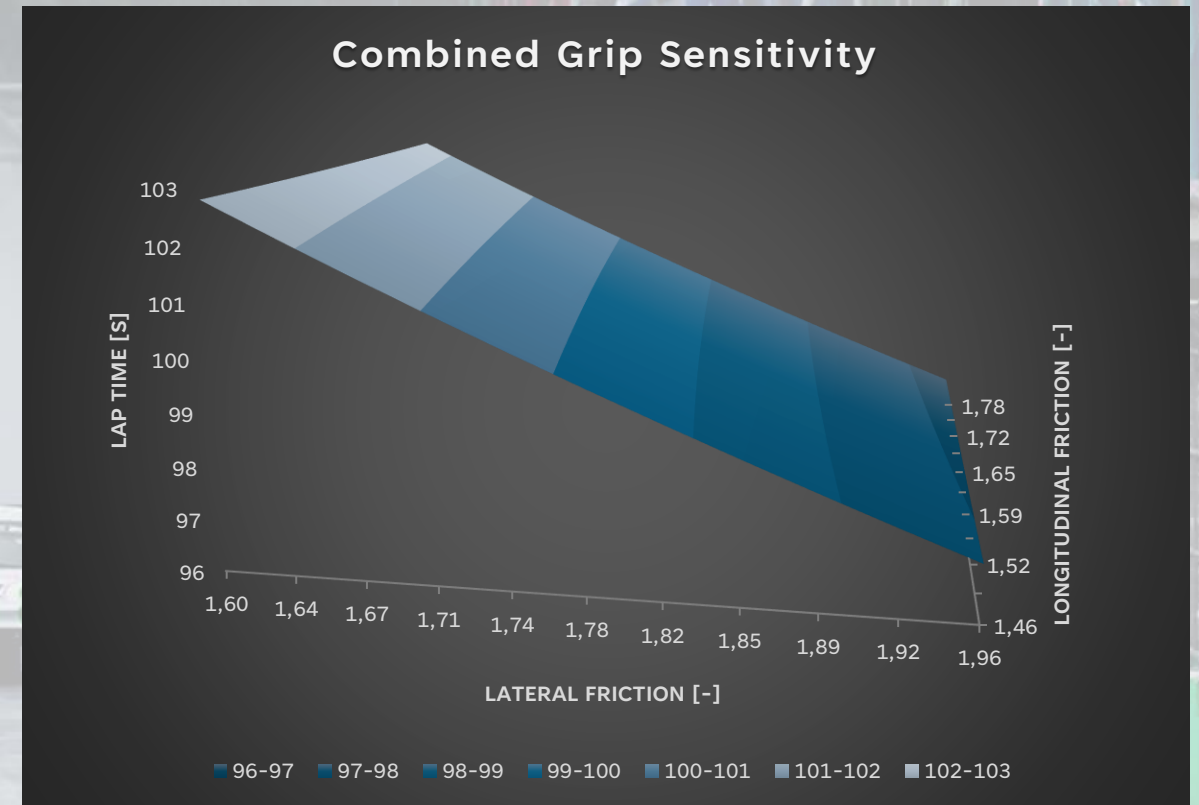


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Parameter Sensitivites:

- **Combined Grip Sensitivity:** Lap time decreases as both lateral and longitudinal friction increase, because higher combined grip allows the car to carry more speed through corners and shorten braking distances. The analysis shows that **lateral grip has the stronger effect**, as cornering limitations dominate the lap around Interlagos.

Lateral Friction	Longitudinal Friction										
	1,60	1,64	1,67	1,71	1,74	1,78	1,82	1,85	1,89	1,92	1,96
	1,46	102,79	102,15	101,53	100,93	100,34	99,76	99,20	98,66	98,12	97,60
	1,49	102,73	102,09	101,47	100,87	100,28	99,71	99,15	98,60	98,07	97,55
	1,52	102,67	102,04	101,42	100,81	100,22	99,65	99,09	98,55	98,01	97,49
	1,56	102,62	101,98	101,36	100,76	100,17	99,60	99,04	98,50	97,96	97,45
	1,59	102,57	101,93	101,31	100,71	100,12	99,55	98,99	98,45	97,92	97,40
	1,62	102,52	101,88	101,26	100,66	100,08	99,50	98,95	98,40	97,87	97,35
	1,65	102,47	101,84	101,22	100,62	100,03	99,46	98,90	98,36	97,83	97,31
	1,68	102,43	101,79	101,18	100,57	99,99	99,42	98,86	98,32	97,79	97,27
1,72	102,39	101,75	101,13	100,53	99,95	99,37	98,82	98,28	97,75	97,23	96,72
1,75	102,35	101,71	101,09	100,49	99,91	99,34	98,78	98,24	97,71	97,19	96,68
1,78	102,31	101,67	101,05	100,45	99,87	99,30	98,74	98,20	97,67	97,15	96,65



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Pre-event context and influencing factors:

- First race of the season → the team is still finding its rhythm, whereas by the middle of the season they are much more in sync and precise
- The track is in Sao Paulo, 5 hour difference to Hungary → Team members need time to adapt
- A Brazilian Superbike event will take place on April 3-5 → impact on track grip levels, and scheduling

Weather:

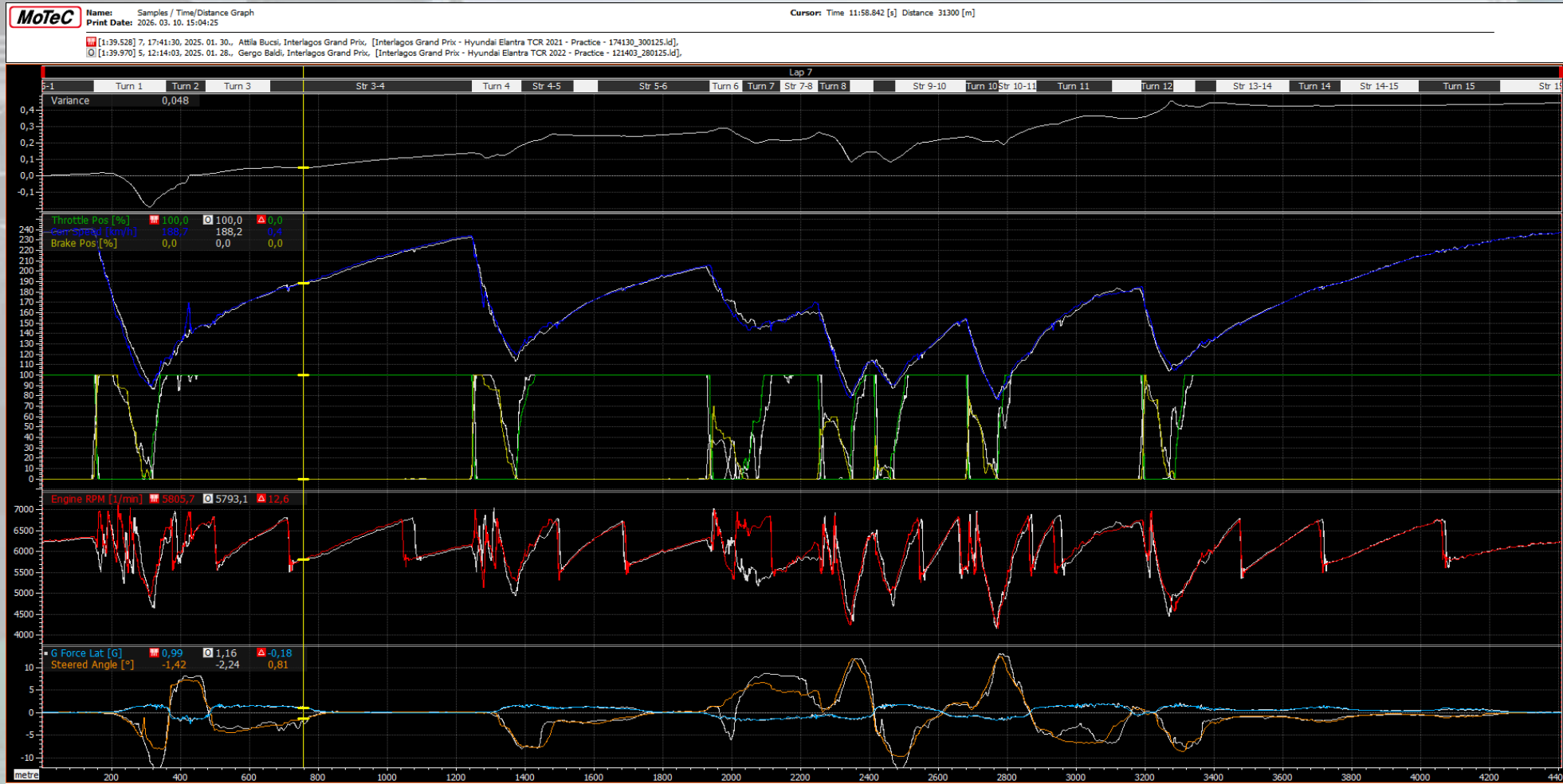
- April sees moderate rainfall and temperatures from **17-27°C**, with **rapidly shifting** weather patterns
- In 2025 the TCR South America grid had rain for the 2nd race, which resulted in a **rolling start** (it was also raining in Q1 and Q2)
- During the afternoon, **shadows** will cover the **odd-numbered grid slots** and later extend to **T7-10 and T14-15**, potentially affecting track temperature

Previous Qualification and Race experiences:

- **Red flag and SC** very likely both in quali and race (From the last 4 previous races, there were 3 accidents in T1)
- **SC** duration: approx. **2-3 laps**
- 2024 WTCR **R1: 14 laps** (3laps behind SC); 2024 WTCR **R2: 18 laps** (2laps behind SC) - Similar to TCRSA 2025 (wet)
- **Slipstream** is really strong in **Str 1**, almost **0.2 seconds faster** with it → 2024 WTCR Q2 Santiago Urrutia
- Highly anticipated track (28 cars) → getting an **ideal lap in Q1 is key**
- In **wet** conditions, **overtaking** is possible **across the entire circuit** → 2025 TCR SA R2, had multiple overtakes

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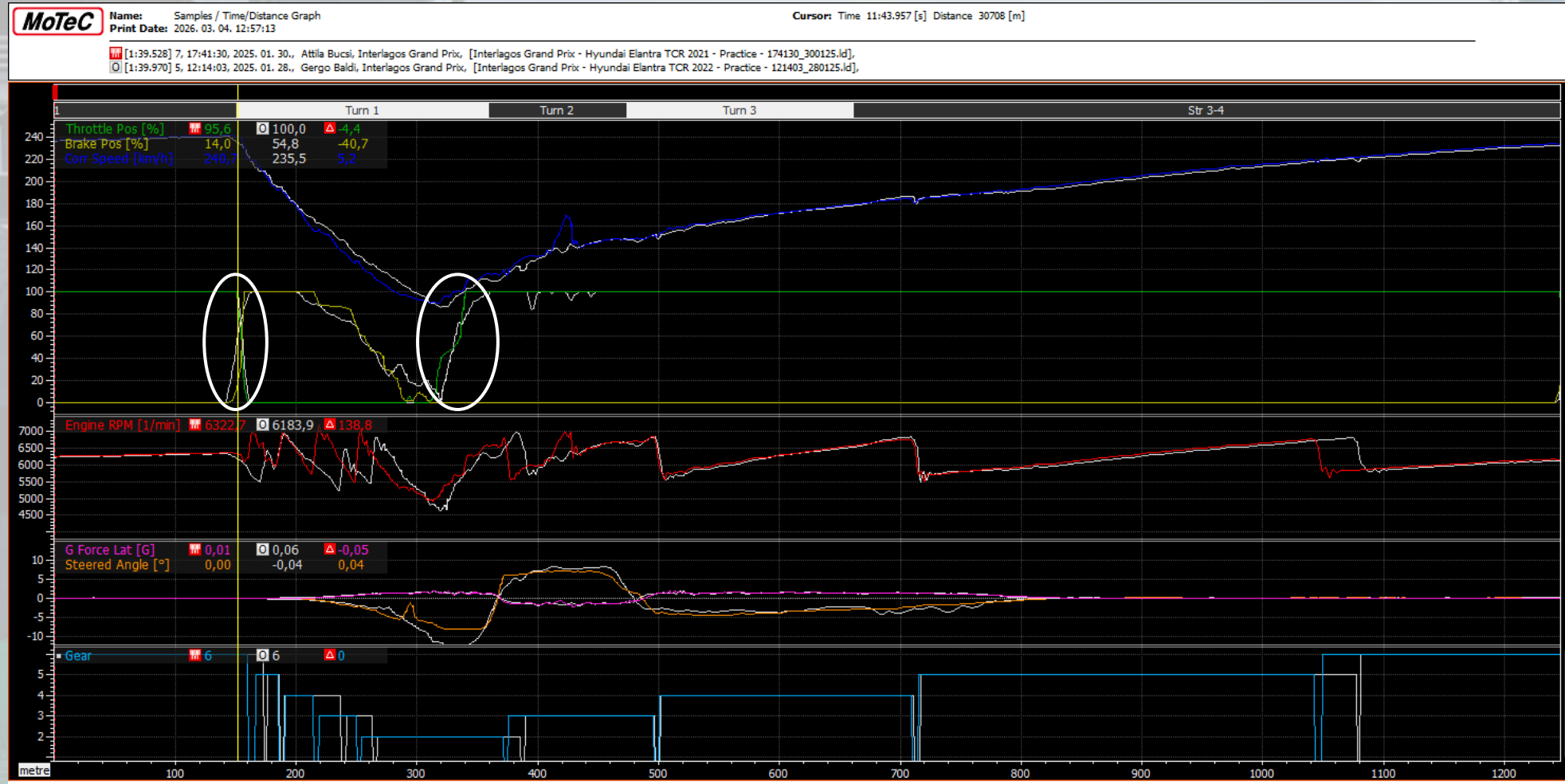
Head to Head driver comparison from simulation data



Attila smoothly builds up a strong lap, and gains a lot through the whole track, expect T1-T2 & T8-T9, where Gergo excels. These two turn combinations are the hardest to maximize.

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Section 1

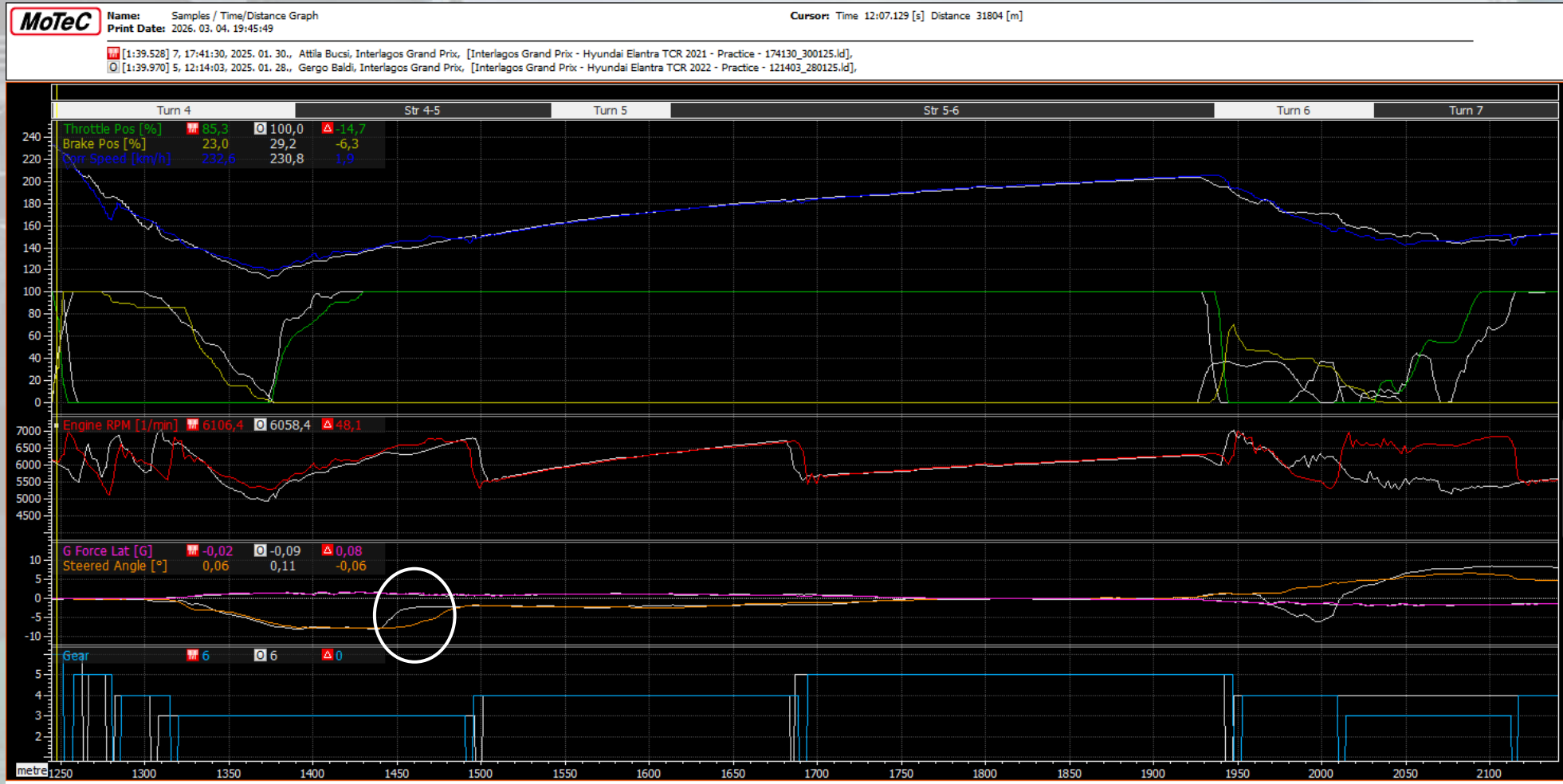


Braking point speed	240,9 km/h
Minimum speed	85,5 km/h
Minimum gear	2
Lowest rpm	4667 1/min
Highest steering angle	-12,82°
Highest Lateral G Force	-2,57 G

T1 is one of the slowest point of the track, high usage of **kerbs** is advised, to maximize your run to **Str 3-4**. Gergo breaks a bit later into T1, but because of that he can't exit the turn as fast as Attila.

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T4-T7

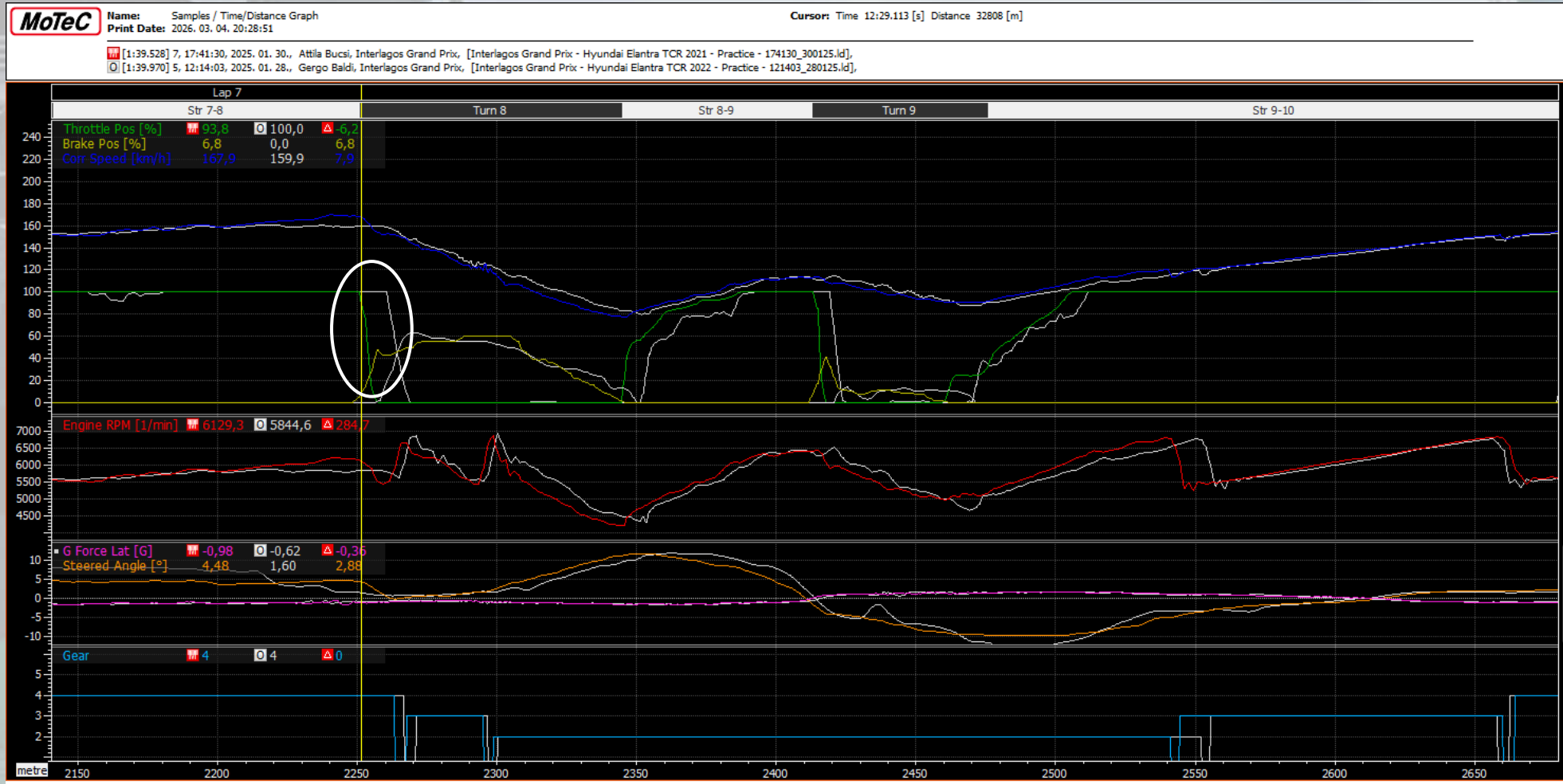


Braking point speed T4/T6-T7	233/205 km/h
Minimum speed T4/T6-T7	113/142 km/h
Minimum gear T4/T6-T7	3/2
Lowest rpm T4/T6-T7	5292/5210 1/min
Highest steering angle	-8/9°
Highest Lateral G Force	1,54/-1,81 G

T7 is an extension of T6, where maintaining the **ideal line is critical**. In wet conditions, this section is a frequent spot for **dive-bombing**. Out of T4, Gergo exits faster, but has to correct the steering angle, and eventually loses speed.

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Str7-8 - Str9-10



Braking point speed T8/T9	168/113 km/h
Minimum speed T8/T9	77/87 km/h
Minimum gear T8/T9	2/2
Lowest rpm T8/T9	4221/4682 1/min
Highest steering angle	11/-13,21°
Highest Lateral G Force	-1,5/1,7 G

T8 is among the **slowest corners** with the **lowest RPM** on the circuit; it is worth evaluating a downshift to a lower gear to optimize performance. In this section Gergo is faster, he brakes later into T8 and can maintain a higher speed.

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T10-Str11-12

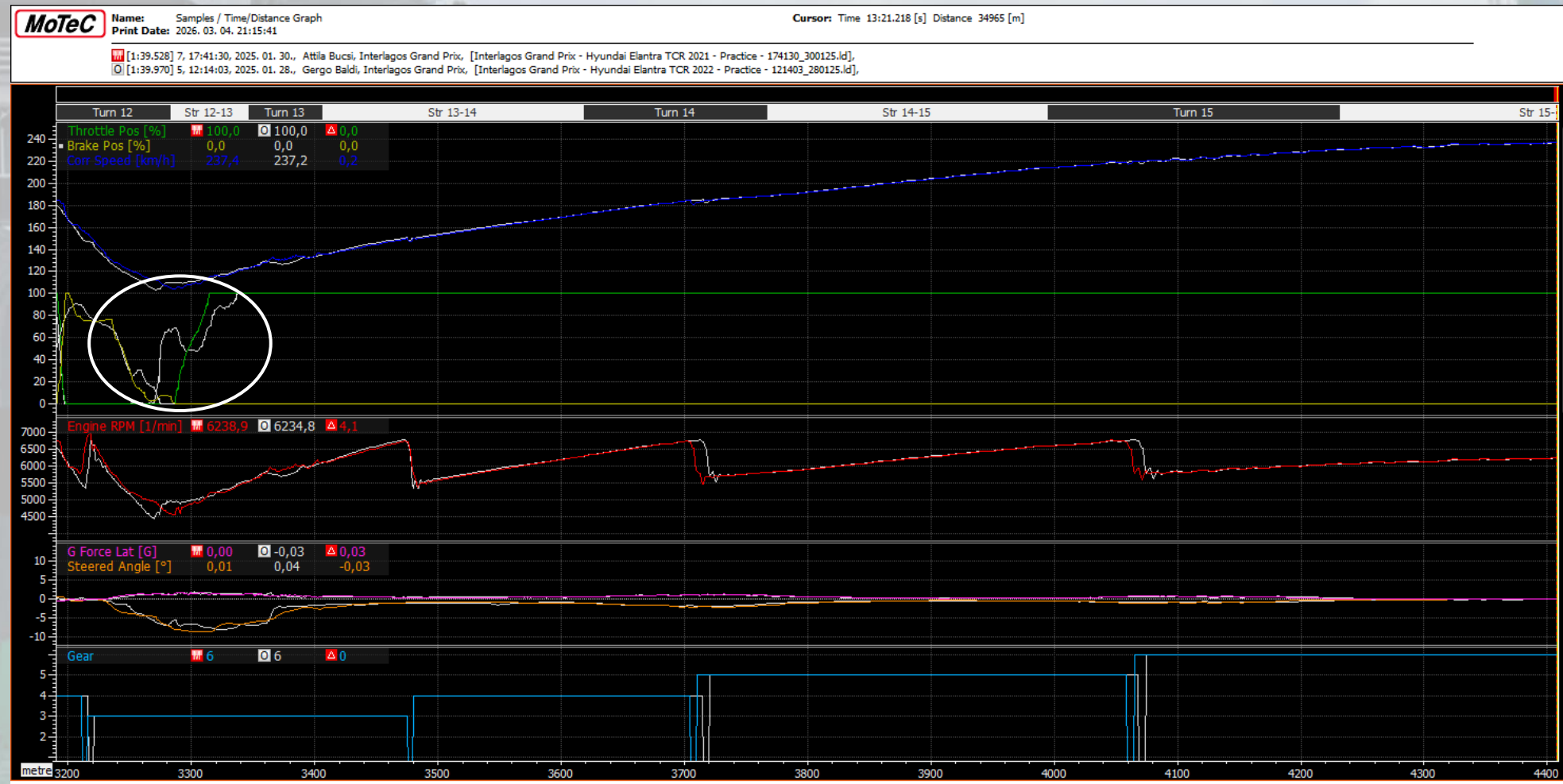


Braking point speed T10	154 km/h
Minimum speed T10	75,6 km/h
Minimum gear T10	2
Lowest rpm T10	4154 1/min
Highest steering angle	12°
Highest Lateral G Force	1,7 G

T10 is the slowest point on the circuit. During FP, it is worth evaluating if a downshift to 1st gear improves mid-corner rotation or exit. T11 is a long, demanding corner. Attila has a better and smoother corner exit from T10.

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Section 3



Braking point speed	184,9 km/h
Minimum speed	103 km/h
Minimum gear	3
Lowest rpm	4423 1/min
Highest steering angle	-8,69°
Highest Lateral G Force	1,74 G

T12 is a critical and demanding corner with significant elevation change, where exit speed is essential to maximise performance along the following long straight; T13–T14–T15 are short, flowing left-hand turns.

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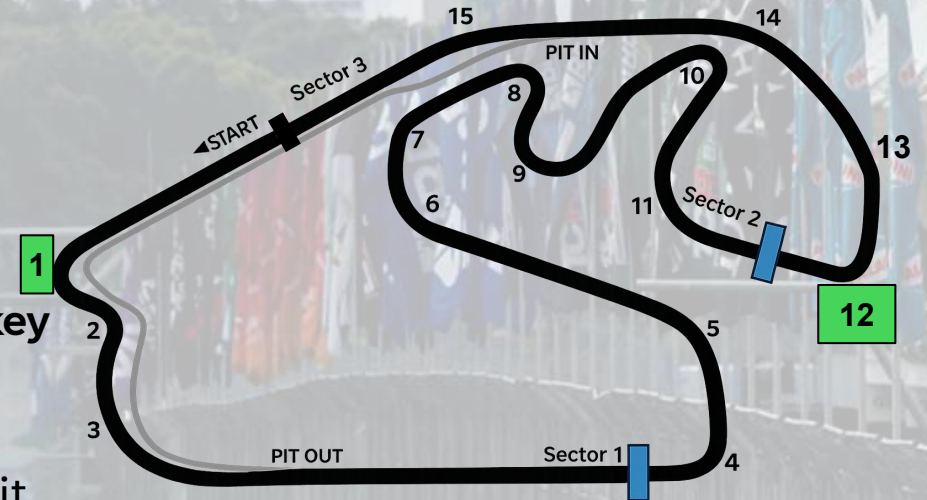
General track information:

- Length: **4309** m, 2nd longest track on calendar
- Elevation: **43** m, 3rd highest on calendar → **Elevation changes are high and important in T1 and in T12**
- Corners: **15** → Left: 10, Right: 5; Ratio → **2:1**
- Track Direction: **Anti** Clockwise
- Expected laptime (dry/wet): **1:42.615/1:53.109**
- Distance from starting line to T1 braking (dry/wet) ≈ **151/157** m

Qualification and Race:

- Highly anticipated track (28 cars) → getting an **ideal lap in Q1 is key**
- **Overtaking** → **T1** and **T4** are the **best** options in dry conditions
- In **wet** conditions, **overtaking** is possible **across the entire circuit**
- **Slipstream** is really strong in **Str 1**, almost **0.2 seconds faster** with it
- **T1** → One of the most important turns on the track, this is where **most overtakes and incidents** happen
- **T12** → With a **good corner exit** you can get a strong slipstream and **overtake**, or you can hold your position
- **Kerb** usage is high, and drivers are encouraged to make full use of them

Race start : High incident risk in T1 (3/4 races). **Stay inside, avoid trouble, and link T1–T2–T3** cleanly to maximize your run onto **Str 3–4**.



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Key Data:

- Q1= 20 min → 1200 s
- Q2=10 min → 600 s
- 6 Tyre set for the weekend (1-1 needed for R1-R2)
- Ideally, the tyre sets used during FP should remain competitive for Qualifying as well.
- Tyre degradation is critical for the front right tyres → data from MoTeC and experiences from previous races
- In lap (dry/wet) $\approx$ 1:50.250 / 2:00.266 → derived from previous race weekends and cross-validated with MoTeC
- Out lap (dry/wet) $\approx$ 1:55.780 / 2:05.422 → derived from previous race weekends and cross-validated with MoTeC
- Pit stop: 90 s
- Speed limit in pit lane: 60 km/h
- Fuel used /lap $\approx$ 2 l → MoTeC data
- Fuel needed in dry conditions according to [Qualification Strategies](#) → Best case: 35 l (Includes 3 l margin)
- Fuel needed in dry conditions according to [Qualification Strategies](#) → Worst case: 37 l (Includes 3 l margin)
- Fuel needed in wet conditions according to [Qualification Strategies](#) → Best case: 35 l (Includes 3 l margin)
- Fuel needed in wet conditions according to [Qualification Strategies](#) → Worst case: 35 l (Includes 3 l margin)
- WTCR 2024 Pole position time (dry): 1:41.840
- WTCR 2024 10th place time (dry): 1:43.234
- TCR SA 2025 Pole position time (wet): 1:51.996
- TCR SA 2025 10th place time (wet): 1:56.458

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Q1 strategy in case of dry conditions:

Q1	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10	L11	L12	L13	Time left (s)*	Tyre sets used	Fuel used (l)
Opt1	out	L1	in	Pit	out	L2	in	Pit	out	L3	L4	in		147,2	3	20
Opt2	out	L1	L2	in	Pit	out	L3	L4	in	Pit	out	L5	in	44,0	3	22
Opt3	out	L1	in	Pit	out	L2	L3	in	Pit	out	L4	L5	in	44,0	3	22
Opt4	out/in	Pit R	out	L1	L2	in	Pit	out/in	Pit R	out	L3	L4	in	58,1	2	20
Opt5	out	abort/in	Pit	out/in	Pit R	out	L1	L2	in	Pit	out	L3	in	42,4	3	20

Opt1: The optimal approach. The driver has three runs on peak-condition tyres and one on a used set to deliver the fastest lap, provided tyre preparation during FP is executed at a high level.

Opt2: No margin for error with limited qualifying time remaining, though up to five timed laps are still possible.

Opt3: Similar to Opt2, the final choice should depend on traffic and track conditions.

Opt4 : If tyre preparation during FP was insufficient and additional conditioning is necessary, while still allowing the driver 4 timed laps.

Opt5: This approach would be necessary if the driver makes an error on a push lap or if an unexpected on-track incident forces the timed lap to be aborted. Worst case scenario.

Q2 strategy in case of dry conditions:

Q2	L1	L2	L3	L4	L5	L6	L7	Time left (s)*	Tyre sets used	Fuel used (l)
Opt1	out	L1	in	Pit	out	L2	in	67,7	2	12
Opt2		out	L1	L2	L3	L4	in	176,3	1	10
Opt3	out	abort/in	Pit	out	L1	L2	in	62,71	2	12

Opt1: The optimal approach. Similar method to Q1 opt1.

Opt2 : Alternative strategy, best applied when targeting P10 to secure reverse-grid position. L3 & L4 are not ideal for strong hot laps.

Opt3: This approach would be necessary if the driver makes an error on a push lap or if an unexpected on-track incident forces the timed lap to be aborted.

*Time left before the start of the last timed lap
|R| Front-rear tyre swap only

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Q1 strategy in case of wet conditions:

Q1	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10	L11	L12	Time left (s)*	Tyre sets used	Fuel used (l)
Opt1	out	L1**	L2	L3	in	Pit	out	L4**	L5	L6	in		167,2	2	20
Opt2	out/in	Pit R	out	L1	L2	in	Pit	out	L3**	L4	L5	in	67,8	2	20
Opt3	out	abort/in	Pit	out	L1	L2	in	Pit	Out	L3	L4	in	56,5	3	20

Opt1: The optimal approach. During wet conditions it is **key to have as many laps as possible**. The driver has six runs on peak-condition tyres to deliver the fastest lap, provided tyre preparation during FP is executed at a high level.

Opt2 : If tyre preparation during FP was insufficient and additional conditioning is necessary, while still allowing the driver 5 timed laps.

Opt3: This approach would be necessary if the driver makes an error on a push lap or if an unexpected on-track incident forces the timed lap to be aborted.

Q2 strategy in case of wet conditions:

Q2	L1	L2	L3	L4	L5	L6	L7	Time left (s)*	Tyre sets used	Fuel used (l)
Opt1	out	L1	in	Pit	out	L2	in	25,9	2	12
Opt2		out	L1	L2	L3	L4	in	131,1	1	12
Opt3	out	abort/in	Pit	out	L1	L2	in	20,5	2	12

Opt1: The optimal approach. Similar method to Q1 Opt1.

Opt2 : *Alternative strategy*, best applied when targeting P10 to secure reverse-grid position. L3 & L4 are not ideal for hot laps.

Opt3: This approach would be necessary if the driver makes an error on a push lap or if an unexpected on-track incident forces the timed lap to be aborted.

*Time left before the start of the last timed lap
** Should be light on tyres/2nd Warm up laps
|R| Front-rear tyre swap only